

HANFORD, WA, USA

WMO: 727840

Lat: 46.567N

Lon: 119.600W

Elev: 223

StdP: 98.67

Time Zone: -8.00 (NAP)

Period: 86-95

WBAN: 94187

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.5	-9.3	-18.1	0.8	-10.2	-13.8	1.2	-6.5	13.4	9.8	11.0	6.8	2.7	20	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	38.2	19.6	36.3	18.6	34.3	17.9	20.1	36.5	19.2	34.4	18.4	32.9	3.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.5	22.3	13.1	9.7	23.1	12.0	9.0	23.3	58.3	36.3	55.0	34.6	52.6	32.8	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.4	8.3	DB	-15.2	40.8	4.7	1.9	-18.6	42.2	-21.4	43.3	-24.0	44.4	-27.4	45.7	(4)
(5)			WB	-15.5	20.8	4.4	2.5	-18.7	22.6	-21.2	24.1	-23.7	25.5	-26.9	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	1.0	3.6	8.3	13.0	17.5	21.7	24.5	24.7	20.0	12.9	5.6	0.3
	DBStd	9.60	4.88	5.37	3.49	3.52	4.41	4.33	4.00	3.66	3.86	4.18	4.44	5.03
	HDD10.0	1006	279	184	72	11	1	0	0	0	0	17	142	301
	HDD18.3	2658	537	413	311	166	71	14	2	2	28	175	381	558
	CDD10.0	2029	1	4	19	99	233	351	450	456	299	106	11	1
	CDD18.3	641	0	0	0	4	46	116	193	199	77	6	0	0
	CDH23.3	8469	0	0	2	78	636	1507	2562	2573	1017	94	0	0
	CDH26.7	4311	0	0	0	15	285	753	1412	1384	444	17	0	0
Wind	WSAvg	3.5	3.0	3.2	3.5	4.0	4.1	4.2	4.2	3.7	3.3	3.0	3.3	2.7
Precipitation	PrecAvg	186	24	14	13	14	15	16	7	5	5	17	23	29
	PrecMax	317	62	29	29	43	36	50	42	25	21	34	46	91
	PrecMin	116	2	0	0	0	0	0	0	0	0	2	0	6
	PrecStd	48	15	8	9	11	10	12	9	6	5	9	13	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.4	22.8	28.7	36.6	39.6	41.1	39.6	35.9	28.8	18.5	13.8
		MCWB	9.8	12.1	11.9	15.8	19.2	20.5	20.9	20.2	18.2	15.7	12.2	8.7
	2%	DB	12.4	15.8	19.7	25.5	33.1	36.0	38.4	37.8	33.7	26.1	16.3	11.4
		MCWB	7.7	9.9	10.7	14.0	17.3	18.7	19.3	19.5	17.2	14.8	10.7	7.2
	5%	DB	10.0	13.3	17.2	23.1	29.9	33.5	36.2	36.0	31.5	23.5	14.3	9.3
		MCWB	6.1	7.9	9.6	12.4	16.0	17.6	18.4	18.7	16.6	13.5	9.5	5.7
	10%	DB	7.5	11.0	15.3	20.8	26.7	31.1	34.1	33.6	29.2	20.8	12.2	6.9
		MCWB	4.5	6.4	8.5	11.4	14.5	16.6	17.8	17.7	15.7	12.0	7.8	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	12.9	12.8	16.6	20.8	20.7	21.1	21.2	18.8	16.7	12.8	9.5
		MCDB	14.5	17.4	20.5	27.1	34.8	38.8	39.5	38.4	34.7	27.4	17.0	12.9
	2%	WB	8.0	10.1	11.5	15.0	17.7	19.1	20.0	20.1	17.8	15.2	11.5	7.6
		MCDB	11.8	15.2	17.6	23.3	31.0	34.8	36.3	35.9	32.2	25.0	15.5	10.8
	5%	WB	6.5	8.1	10.4	13.4	16.3	18.0	19.0	19.1	16.9	14.0	10.0	6.0
		MCDB	9.7	12.4	16.3	21.6	28.3	32.1	34.4	34.0	30.4	21.9	13.8	8.7
	10%	WB	4.9	6.9	9.1	12.0	15.1	17.0	18.2	18.3	16.1	12.7	8.3	4.5
		MCDB	7.0	10.3	14.4	19.5	25.8	29.7	32.6	32.3	28.5	19.6	11.7	6.7
Mean Daily Temperature Range	5% DB	MDBR	7.1	9.5	12.2	13.3	14.2	14.5	15.2	15.6	16.0	13.4	8.7	6.2
		MCDBR	10.3	13.0	15.1	16.7	18.2	17.2	17.5	17.6	19.2	16.1	9.9	9.8
		MCWBR	6.4	7.9	8.4	8.1	7.2	7.0	6.4	6.6	7.8	7.6	5.9	6.5
	5% WB	MCDBR	10.0	11.5	13.8	15.1	17.0	16.3	16.8	16.4	17.8	15.4	9.3	9.2
		MCWBR	6.4	7.2	8.0	7.4	7.1	6.8	6.4	6.4	7.3	7.3	5.9	6.3
Clear-Sky Solar Irradiance	taub		0.286	0.290	0.299	0.331	0.346	0.346	0.347	0.344	0.331	0.309	0.293	0.282
	taud		2.548	2.566	2.563	2.439	2.415	2.440	2.442	2.443	2.476	2.538	2.544	2.539
	Ebn at Noon		829	900	938	926	919	918	912	902	885	860	810	790
	Edh at Noon		62	73	85	106	112	110	109	104	92	74	61	56
All-Sky Solar Radiation	RadAvg		1.14	2.25	3.56	5.29	6.41	7.16	7.65	6.42	4.75	2.93	1.51	0.95
	RadStd		0.15	0.29	0.24	0.30	0.37	0.35	0.26	0.35	0.26	0.23	0.14	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	+42

Nomenclature: See separate page